Regi Kusumaatmadja

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Education

VU Amsterdam and Tinbergen Institute, Ph.D. in Economics, Expected Completion: 2026.
Tinbergen Institute, M.Phil. in Economics, 2021.
VU Amsterdam, M.Sc. in Economics, 2018.
University of Indonesia, B.Sc. in Physics, 2016 (Best Graduate).

Letter Writers

prof. dr. José Luis Moraga-González
VU Amsterdam & Télécom Paris
Tinbergen Institute, CEPR, CESifo
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prof. dr. Sabien Dobbelaere
VU Amsterdam
Tinbergen Institute, IZA
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prof. dr. Mark J. Roberts
Pennsylvania State University
NBER, ZEW
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prof. dr. Eric J. Bartelsman (placement dir.)
VU Amsterdam
Tinbergen Institute, IZA
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Visiting Position

2026.05 - 2026.06, CUNEF Universidad Madrid, invited by **dr. Costanza Cincotta**
2023.01 - 2023.06, the Pennsylvania State University, invited by **prof. dr. Mark J. Roberts**

Research and Teaching Areas

Primary Fields: Industrial Organization, Economics of Innovation
Secondary fields: Applied Econometrics, Applied Microeconomics

Research in Progress

Internal and External R&D: An analysis of costs and benefits

This paper analyzes the costs and benefits of internal and external R&D activities. Using Dutch production and innovation surveys between 2000 and 2020, focusing on the ICT industry, I document an increasing share of firms performing R&D and the four-way mix of in-house research, contracted-out research, both, and neither. To rationalize these findings, I build and estimate a dynamic discrete-choice model of R&D with mode-specific investment costs. The cost of contracted-out R&D is much higher than the cost of in-house R&D, which explains the small share of firms that contract out alone. Doing both is cheaper than the sum of the two costs: in-house R&D offsets almost all of the extra cost of going outside. I read that extra cost as a hold-up problem with the partner. The offset is absorptive capacity: a firm that already does in-house R&D keeps more of the collaboration, so the two activities cost less together than apart. Productivity gains are similar across modes, so the mix of activities is driven by these costs. A uniform subsidy that does not favor a mode raises the share of R&D-active firms and firm value by more than a subsidy aimed only at in-house R&D.

Presentation: JEI - Gran Canaria (2022), Ph.D.-EVS (2022), IO/Applied Micro Brownbag - Penn State (2023), VU Amsterdam Internal Seminar (2023), EARIE - Rome (2023), Tinbergen Institute Ph.D. Seminar (2023), DKIIS - Valencia (2023), CAED - Penn State (2024), SED Meeting - Barcelona (2024), CEPR Workshop: Trade, Geography, and Industrial Organization - Erasmus Rotterdam (2024, poster), APIOC - Seoul (2024), CEPR Symposium - Paris (2024, poster), EEA-ESEM - Bordeaux (2025), CEPR IO Virtual Gathering (2025)

Entry into Innovation Areas

This paper investigates the multi-area entry decisions of publicly listed firms in the United States electronics industry from 1990 to 2019. I apply Latent Dirichlet Allocation to over 630,000 patent titles and abstracts to endogenously define twenty innovation areas and merge these with market-implied patent valuations. The data reveal three empirical patterns. First, innovation occurs across multiple domains simultaneously. Second, the number of active competitors varies across areas. Third, the industry focus shifted from physical hardware to connectivity technologies. To explain these patterns, I estimate a static simultaneous-move game of firm entry using a partial identification approach with moment inequalities, which accommodates multiple equilibria and the combinatorial choice set. The 99 percent confidence sets support four conclusions. First, firms enter areas where expected patent valuations are higher. Second, holding those valuations fixed, additional rivals reduce expected profit. Third, firms also benefit from occupying more areas at once. Fourth, a large per-area entry cost, together with rival congestion, still limits how many areas a firm enters. Starting from observed portfolios, between 13 and 16 percent of firm-years already have at least one extra area that looks worth occupying at the estimated costs. A 25 percent reduction in the per-area cost raises that share to between 22 and 27 percent, and a 50 percent reduction raises it to between 33 and 41 percent. A 50 percent reduction in how costly rivals are raises that share to between 15 and 21 percent. In a case study of targeted cost reductions, entry rises in semiconductor areas such as "Semiconductor Memory Architecture" and "Semiconductor Lithography and Fabrication Processes." The offset is lower entry in other fields, including "Wireless Base-Station and Mobile Communications."

Presentation: VU Amsterdam Internal Seminar (2025)

The Impact of Government Financial Assistance on the Scope and Direction of Firm Innovation, with C. Cincotta.

Governments spend hundreds of billions of dollars each year to support private-sector innovation, but it is unclear whether this money expands existing research, redirects it toward new technologies, or reallocates innovation away from other firms. To address this question, we develop a two-stage model in which firms first choose an R&D portfolio and then compete in prices à la Salop. Financially constrained firms underinvest, and even unconstrained firms under-provide quality because product-market competition leaves a gap between private and social returns. Liquidity assistance can relax the cash constraint, while the quality gap remains. Cheaper cash can expand how much a recipient patents and how many technology areas it covers. If the firm simply scales up the research it already conducts, the composition of that research does not change. Assistance that is targeted toward particular activities can redirect the portfolio. Nearby rivals may expand through knowledge pooling or contract through business stealing. We take these implications to a panel of U.S. public firms from 2000 to 2021, combining Good Jobs First awards with patent data. Scope measures how much a firm patents and how many technology areas it covers. Direction measures whether those patents align with frontier or peer technologies. To estimate causal effects, we first reweight untreated firms so that they match recipients on pre-treatment characteristics, and we then estimate a stacked difference-in-differences. Our model implies that distant rivals provide a better control group for the treated firm's own response. Nearby rivals may themselves be affected by business stealing or knowledge pooling, so comparing the two groups traces how those nearby rivals respond. We implement this design using both industry codes and text-based product-market overlap. Distant rivals provide the preferred reading of the treated firm's own response. Financial assistance expands the scale of recipients' patent portfolios. Relative to distant industry controls, intensive scope rises by about 10.5 percent after an award. Relative to firms with little product-market overlap, the same contrast is about 3 percent. Direction shifts toward the industry's most-cited technologies, and the shift is strongest for cash grants. When nearby means high text-based product-market overlap and distant means little or none, the product-market spills are indistinguishable from zero, so those nearby rivals' patenting scale does not move. When nearby means the same two-digit industry, the close estimate on scale is smaller than the distant estimate, which means industry-code neighbors expand as well, the knowledge-pooling case.

Mergers and Innovation: An Exploration of Scope and Direction of Innovation, with S. Dobbelaere and J. L. Moraga-Gonzalez.

How do mergers change the scale and composition of firm innovation? We study U.S. public mergers from 1980 to 2020 using combined acquirer–target patent portfolios. We measure *scope*, the scale of the joint portfolio, and *direction*, its alignment with highly cited patents. A compact portfolio model isolates three forces: product-market business stealing, technological knowledge pooling, and fixed costs of activating innovation lines. Business stealing can reshape private returns where the firms already compete. Knowledge pooling raises the productivity of post-merger R&D where technologies are related. Activation lets the merged firm open innovation lines that were not worth running before the merger. The model also speaks to reallocation: how post-merger effort is split between lines already shared by both firms and lines unique to one. The model leaves the average change in that split unsigned, and it leaves the average change in direction unsigned as well. We compare treated merger pairs to matched control pairs. Empirically, on the matched sample, joint-portfolio scope is larger after mergers than for matched controls. The difference is substantially larger when pre-merger technologies are more related, especially among non-horizontal deals. We find no strong evidence of reallocation. Innovation activity expands after mergers, without a clear rebalancing of activity between technology classes shared by both acquirer and target and classes unique to one. On direction, joint portfolios show higher alignment with highly cited patents relative to matched controls. Acquirer-only R&D measures overstate post-merger input growth relative to joint-entity measurement.

Presentation: VU Amsterdam Internal Seminar (2024), Tinbergen Institute Workshop in Economics of Innovation (2025)

Teaching Experience

2021-2025 TA, VU Amsterdam, Introduction to Econometrics (MSc). Eval: 4.3/5.0
2021-2025 TA, VU Amsterdam, Applied Econometrics (MSc). Eval: 4.7/5.0
2022-2025 Supervisor, VU Amsterdam, Research Project and Thesis (BSc + MSc).
2021-2024 Instructor, VU Amsterdam, Math, Statistics, Micro, and R (Refresher MSc).
2022 TA, VU Amsterdam, Microeconomics I (BSc).
2021-2025 Guest Lecturer, VU Amsterdam, Urban Economic Challenges & Policies (MSc).

Relevant Experience

2023, EARIE 1st Summer School on Entry Model, LUISS - Rome.
2022, Econometric Society's Dynamic Structural Econometrics Summer School, MIT.
2019, Research on Productivity, Trade, and Growth Summer School, Tinbergen Institute.
2020-2021, RA for Eric J. Bartelsman and Sabien Dobbelaere, VU Amsterdam, MICROPROD project.

2018-2019, Research Associate, Institute for Economic & Social Research, University of Indonesia (LPEM FEB UI).

Scholarships, Honors, and Awards

2025, Excellence in Teaching Award, VU Amsterdam.
2023, Research Visit Grant, VU Amsterdam (monetary grant).
2019-2021, Tinbergen Institute - Full Scholarship (tuition & living cost).
2017, VU Amsterdam Fellowship Programme (tuition).
2017, Holland Scholarship Programme (living cost).
2016, Best Graduate, Department of Physics, University of Indonesia (monetary prize).
2016, Top 1 (one) Percent, Faculty of Mathematics and Natural Sciences, University of Indonesia.
2014-2016, XL-Axiata Future Leaders Scholarship (leadership programme & in-kind benefits).
2014-2015, Karya Salemba Empat Scholarship (tuition & living cost).

Publications

Evaluating the congestion-reducing effects of road rationing policy: Evidence from Jakarta's odd-even policy. *Case Studies on Transport Policy* (2025), with M. Halley Yudhistira, Y. Sofiyandi, and M. Firman Hidayat. (previously circulated as Does Traffic Management Matter? Evaluating Congestion Effect of Odd-Even Policy in Jakarta)

Theoretical exploration of optical response of Fe_3O_4-reduced graphene oxide nano-particle system within dynamical mean-field theory. *IOP Conf. Ser.: Mater. Sci. Eng.* (2017), with M.A. Majidi, A.D. Fauzi, W.Y. Phan, A. Taufik, R. Saleh, and A. Rusydi.

Policy Work

The sky in blues: on the recent development of Indonesian airlines industry. *LPEM FEB UI WP* (2019), with C. Nuryakin, N. W. G. Massiw, A. U. Lumbanraja, S. Hambali, and D. Anindita

Miscellaneous

Language: Indonesian (native), English (fluent), Dutch (basic-intermediate), Arabic (basic-intermediate)

Software: R, Python, Rust

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